

Repolyn — chemical recycling that returns PET to virgin-grade

The Problem

Mechanical recycling degrades PET. Each cycle shortens the polymer chains, colour accumulates, and after two or three passes the material is only fit for fibre or for landfill. This is why a bottle is rarely recycled into a bottle, and why brands committed to recycled content are competing for a supply that does not physically exist in the volumes they have promised.

Coloured PET, multilayer packaging and textile-blend polyester are not recycled at all in any meaningful volume. They are the majority of what is produced.

Our Solution

Repolyn depolymerises PET back to its monomers — purified terephthalic acid and ethylene glycol — at 180°C and atmospheric pressure, using a solvent system we developed and a catalyst that is recovered and reused.

The output is chemically identical to virgin feedstock, because it is the same molecule. It can be repolymerised to food-contact grade without a regulatory exception, and it accepts feedstock that mechanical recycling rejects: coloured, multilayer, textile blends, and material contaminated to roughly 12% by weight.

Our pilot line in Switzerland has run 340 tonnes since commissioning, with a monomer yield of 91% and a purity that has cleared food-contact testing on every batch submitted.

The Market

Global PET production is around 30 million tonnes a year. Regulation is the demand driver and it is already law: EU packaging rules require 30% recycled content in PET bottles from 2030, and similar mandates exist in eleven other jurisdictions.

The recycled PET supply gap against those mandates is estimated at 6 to 9 million tonnes a year. That gap is the market.

Competition

Four chemical recycling companies have announced commercial plants; two have commissioned them. Enzymatic approaches are further behind on throughput but ahead on energy use.

Our position is process cost per tonne at atmospheric pressure, which removes the pressure-vessel capital that sits on most competing designs.

The Team

Dr. Elin Rask spent eleven years in polymer chemistry at a materials research institute and holds four patents on the catalyst system. Anders Falk built and commissioned two specialty chemical plants and ran operations at one of them for six years.

Thirty-one people, nineteen of them chemists and process engineers.

Traction

- pilot line commissioned, 340 tonnes processed

- 91% monomer yield, sustained across the last 60 batches
- food-contact certification cleared on output
- offtake agreements signed with two packaging converters, totalling 4,000 tonnes a year on commissioning of the first commercial line
- one brand partnership in trial

Business Model

We sell monomer by the tonne. Our current all-in production cost at pilot scale is €1,410 a tonne. At the 50,000-tonne commercial scale the model projects €890.

Virgin PET feedstock trades between €760 and €1,050 a tonne depending on oil.

So the honest position is this: at commercial scale we are at parity with virgin at the top of its range and above it at the bottom. The gap is closed by the regulatory mandate, by the recycled-content premium brands currently pay — €180 to €340 a tonne — and by a carbon price we do not control. Our offtake agreements are priced against that premium, not against virgin.

Whether that premium persists once supply arrives is the question we cannot answer, and it is the question that decides this business.

Go-To-Market

Direct offtake with packaging converters and brand owners, on multi-year contracts signed before plant construction — which is how project finance for this kind of asset becomes available at all.

Funding

We are raising €23M of equity alongside €40M of project debt to build the first 50,000-tonne line, targeting commissioning in 2028.